

GHARDAIA NOUMERAT, Algeria

WMO: 605660

Lat: 32.384N

Lon: 3.794E

Elev: 461

StdP: 95.91

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	5.0	-9.8	1.7	14.5	-7.8	2.1	15.7	14.5	11.8	12.7	11.9	2.5	330	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	43.8	21.0	42.6	20.7	41.2	20.3	23.0	37.7	22.1	37.1	21.4	36.3	4.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.6	26.7	17.8	13.5	26.9	16.6	12.5	27.4	70.0	37.5	66.4	36.7	63.9	36.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
12.1	10.4	9.1	DB	1.8	45.3	1.1	0.9	1.0	46.0	0.3	46.5	-0.4	47.0	-1.2	47.7	
			WB	-0.8	24.5	1.1	1.7	-1.6	25.7	-2.3	26.7	-2.9	27.7	-3.7	29.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	11.7	13.3	17.1	21.0	26.1	31.0	34.7	33.8	28.9	23.3	16.6	12.6
	DBStd	8.63	2.68	3.19	3.63	3.63	3.68	3.26	2.59	2.51	3.56	3.38	3.15	2.71
	HDD10.0	27	13	6	1	0	0	0	0	0	0	0	1	7
	HDD18.3	684	206	144	67	13	1	0	0	0	2	71	180	
	CDD10.0	4611	66	98	221	330	499	632	767	737	566	412	197	87
	CDD18.3	2227	1	3	29	93	242	382	508	478	317	156	18	1
	CDH23.3	32732	2	29	285	989	3035	5882	8736	7916	4273	1459	120	6
	CDH26.7	20733	0	6	74	392	1610	3786	6290	5494	2466	594	21	0
Wind	WSAvg	3.6	3.5	3.9	4.2	4.5	4.4	4.0	3.4	3.3	3.3	2.9	3.1	3.4
Precipitation	PrecAvg	59	8	3	9	6	3	3	1	2	6	5	7	6
	PrecMax	132	67	22	57	53	20	18	17	21	43	29	31	40
	PrecMin	19	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	24	13	5	13	9	5	4	3	5	9	7	7	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	26.5	31.9	35.9	40.6	43.3	45.5	44.4	41.6	36.8	29.1	23.9
		MCWB	11.7	13.4	15.5	17.3	18.7	20.7	21.5	21.4	20.0	18.9	15.6	13.6
	2%	DB	20.8	23.3	28.7	33.1	38.1	42.0	44.2	43.1	39.6	34.2	26.5	21.2
		MCWB	11.2	11.8	14.0	15.7	18.3	20.3	21.1	21.0	19.5	17.6	14.8	12.2
	5%	DB	18.9	21.4	26.2	30.9	36.1	40.5	43.1	42.0	38.0	32.0	24.3	19.4
		MCWB	10.3	11.2	13.2	15.1	17.5	19.7	20.7	20.8	19.4	17.2	14.1	11.4
	10%	DB	17.1	19.5	24.3	28.9	34.0	38.9	41.9	40.6	36.1	30.0	22.6	17.9
		MCWB	9.5	10.4	12.4	14.7	17.0	19.2	20.5	20.4	19.4	17.0	13.4	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	15.4	16.9	18.7	20.7	23.2	24.2	24.8	22.8	21.2	18.3	15.0
		MCDB	20.3	23.1	25.9	29.5	35.0	38.6	39.9	39.6	33.3	28.6	25.4	21.1
	2%	WB	11.9	13.2	15.5	17.2	19.2	21.5	22.8	23.1	21.7	19.8	16.6	13.3
		MCDB	18.7	21.6	25.7	27.8	33.3	37.2	39.3	37.7	32.3	27.3	23.4	19.0
	5%	WB	10.8	11.8	14.2	16.1	18.5	20.6	21.8	22.0	21.0	19.0	15.2	12.2
		MCDB	17.4	19.7	24.1	27.3	32.5	36.6	39.3	37.0	32.3	27.5	22.5	17.9
	10%	WB	10.0	10.8	13.1	15.2	17.7	19.8	21.2	21.3	20.4	18.2	14.1	11.3
		MCDB	16.2	18.4	22.8	26.5	31.5	35.8	38.5	36.4	31.9	27.0	21.2	17.0
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.4	11.3	11.9	12.1	12.4	12.5	12.0	11.4	10.8	10.1	9.7
		MCDBR	12.1	12.6	13.8	14.5	14.0	14.0	13.4	13.2	13.3	13.0	12.0	11.6
		MCWBR	6.2	6.4	6.1	5.9	5.5	5.0	4.7	4.8	4.5	5.0	5.6	5.9
	5% WB	MCDBR	10.9	11.4	12.5	12.4	12.4	12.5	12.4	12.0	11.6	10.6	10.6	10.1
	MCWBR	6.0	6.3	6.0	5.9	5.4	5.2	5.0	4.7	4.5	4.4	5.5	5.5	
Clear-Sky Solar Irradiance	taub	0.315	0.345	0.438	0.481	0.537	0.569	0.619	0.584	0.549	0.469	0.370	0.330	
	taud	2.343	2.241	1.934	1.850	1.740	1.682	1.600	1.677	1.754	1.920	2.178	2.306	
	Ebn at Noon	900	903	835	816	772	745	705	722	725	754	819	857	
	Edh at Noon	100	123	181	205	232	245	264	241	214	168	118	99	
All-Sky Solar Radiation	RadAvg	3.70	4.65	5.82	6.87	7.41	7.89	7.76	7.05	5.88	4.82	3.84	3.32	
	RadStd	0.18	0.20	0.25	0.27	0.23	0.17	0.14	0.20	0.20	0.24	0.16	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+0.42	N/A	-2.26	N/A	-0.91	-0.62	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.34	N/A	-2.26	N/A	-0.91	-0.62	N/A	N/A	N/A

Nomenclature: See separate page